

AI Factory Command Center

From factory signals to a human-reviewed decision.

AI Factory 2.0 · Bearing manufacturing demonstration

All included factory data and manuals are synthetic.

The factory problem

Rising vibration. More surface defects. A production decision.

- Detect emerging machine risk before the next six hours.
- Inspect product quality and maintenance context.
- Compare the cost of three responses before a supervisor decides.

Five input modalities

Production CSV

Sensor history

Product image

Maintenance note

PDF / text SOP

6 machines · 4,320 sensor observations before injected duplicates · 920 independently seeded images.

Upload controls accept replacement evidence. Image predictions are only validated on the synthetic generator.

One integrated architecture

Inputs + cleaning → Maintenance + vision models → Knowledge retrieval

Structured agents → Digital twin → Supervisor → PDF + audit

MLflow records experiments, metrics, artifacts and the selected model version.

Prevent leakage before training

- Remove duplicate machine timestamps; coerce types and invalid values.
- Forward-fill within each machine; fit remaining imputation on training only.
- Join production records; create rolling mean and variation features.
- Chronological 60 / 20 / 20 split with six-hour purges.
- Retain plausible high readings: they may be the fault signal.

Measured model results

Model	Task	Precision	Recall	F1	AUC
logistic	maintenance	0.875	0.791	0.831	0.947
random_forest	maintenance	0.874	0.844	0.859	0.944
ann	maintenance	0.860	0.844	0.852	0.949
cnv	vision	1.000	0.975	0.987	1.000

Selected maintenance model: **random_forest**, registered version 1. Selection uses validation F1.

Held-out synthetic test set. Maintenance and vision are different prediction tasks; their metrics are not a direct contest.

Show why the models responded

- Global permutation importance measures the Random Forest's reliance on features.
- Local sensitivity replaces each feature with its training median.
- Grad-CAM highlights regions contributing to the CNN's predicted class.

These explain model behavior. They do not establish physical causes or calibrated confidence.

Retrieve before explaining

Symptoms + signals → TF-IDF section retrieval → Cited evidence → Optional LLM

Demonstrate a manual-specific question, then inspect the retrieved section. Select Local Qwen to generate an explanation without paid API credits.

Offline summaries are explicitly extractive. Verified local case: without retrieval the model invented 5% and 1.4; with retrieval it returned the manual values 70% and 0.55. One case does not prove general accuracy.

Four agents exchange evidence

- **Maintenance:** model probability, signals and note extraction.
- **Vision:** defect score and quality-review flag.
- **Knowledge:** retrieved paragraphs and source names.
- **Planner:** scenario comparison and pending recommendation.

Deterministic modular orchestration with visible structured messages; no claim of autonomous reasoning agents.

Simulate the operational choices

Continue Reduce load to 70% Stop for 2h maintenance

- Compare good units, downtime, failure risk and expected cost.
- Convert the six-hour prediction to an eight-hour horizon using an assumed constant hazard.
- Use observed production rejects for yield; image probability is not a batch defect rate.

Hypothetical economics and action effects. This is a simplified digital twin.

Human authority is part of the product

Approve. Reject. Modify.

- Save the decision and evidence snapshot in SQLite.
- Require a reason when modifying the action.
- Download the report with an explicit supervisor status.

Confirmed outcomes can train a candidate. Chronological evaluation and minimum-quality gates protect the active model; promotion is never automatic.

Live demonstration + next steps

1. Choose M-01 and observation 610; analyze the defective sample.
2. Inspect predictions, explanations, sources and agent messages.
3. Compare scenarios; reject or modify the recommendation.
4. Download the PDF and show MLflow experiments.

Before real deployment: real datasets, calibration, external validation, verified SOPs and controlled candidate promotion.